Lec-10
22/06/26 10:15 AM

Recursion

fun calling itself

① fun calling
A ( ) {
B ( )
}

② fun calling itself
A ( ) {
A ( )
}

PMI

(i) $\sum_{i=1}^{i=N} i = 1+2+3+\dots+N = \frac{N(N+1)}{2}$

$\sum_{i=1}^{i=1} i = 1 \rightarrow \frac{1\times(1+1)}{2}$

$\sum_{i=1}^{i=2} i = 1+2 = \frac{2\times(2+1)}{2}$

(ii) $\sum_{i=1}^{i=K} i = 1+2+3+\dots+K = \frac{K(K+1)}{2}$ ✓

(iii) $\sum_{i=1}^{i=K+1} i = \underbrace{1+2+3+\dots+K}+(K+1) = \frac{(K+1)(K+2)}{2}$

$\frac{K(K+1)}{2} + K+1 = \frac{(K+1)(K+2)}{2}$

$(K+1)\left[\frac{K}{2}+1\right]$

$\frac{(K+1)(K+2)}{2}$

① Smaller → input ✓
② N=K correct ✓
③ N=K+1 ✓

Recursion

Recursion [illegible] यह [illegible]
L2R ♡ है।

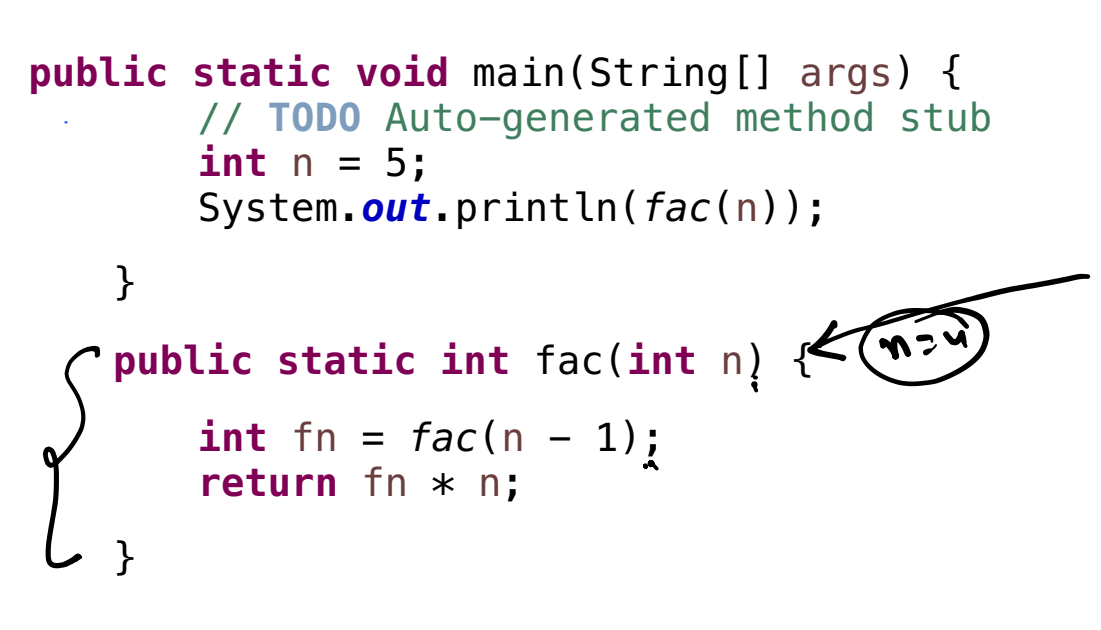
N! = $1\times2\times3\times4\times5\cdots\times N$

(N-1)! → 2nd Step → PMI

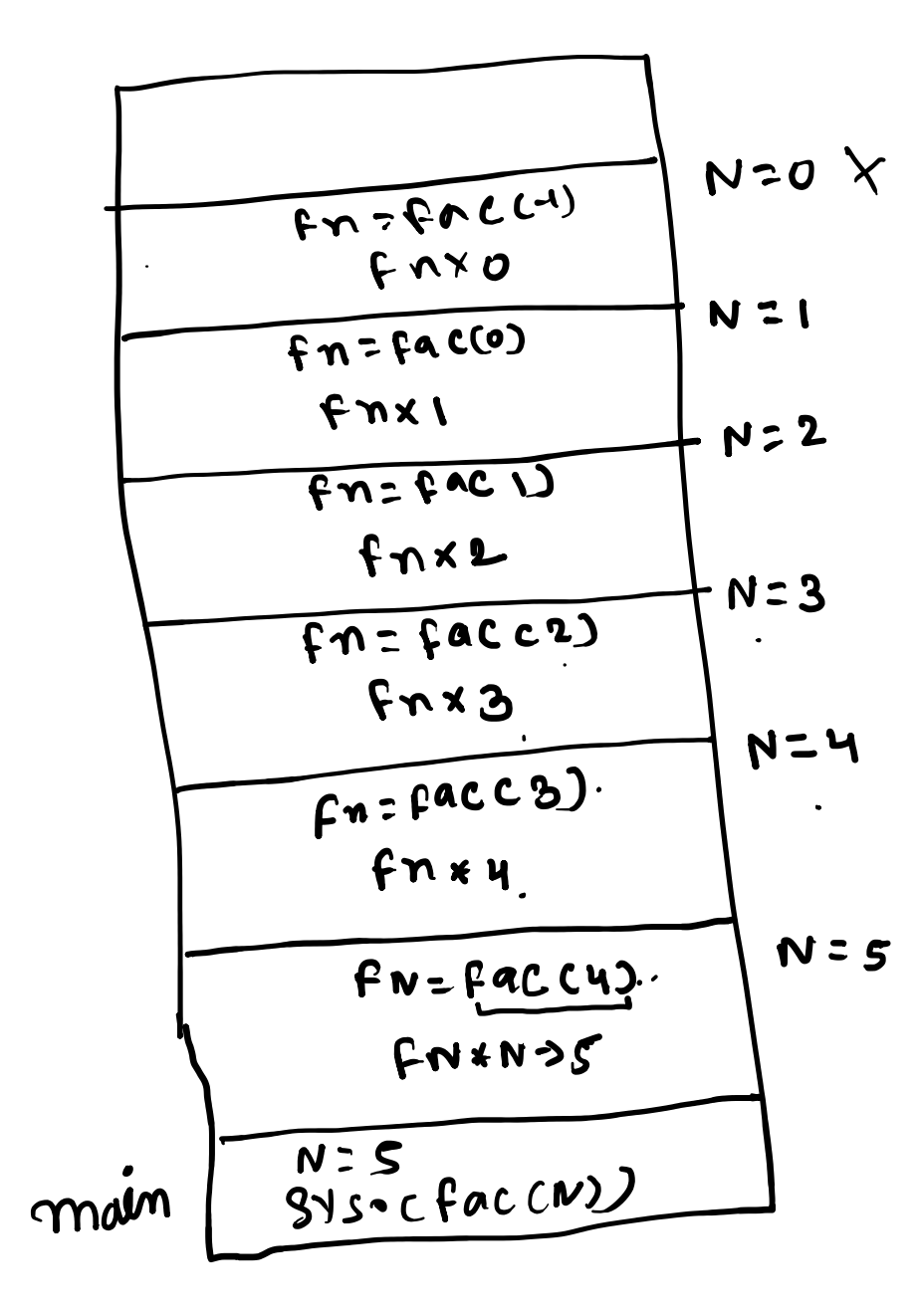
ans — Self Work → 3rd PMI

a^n = $(a^{n-1}) \times a$

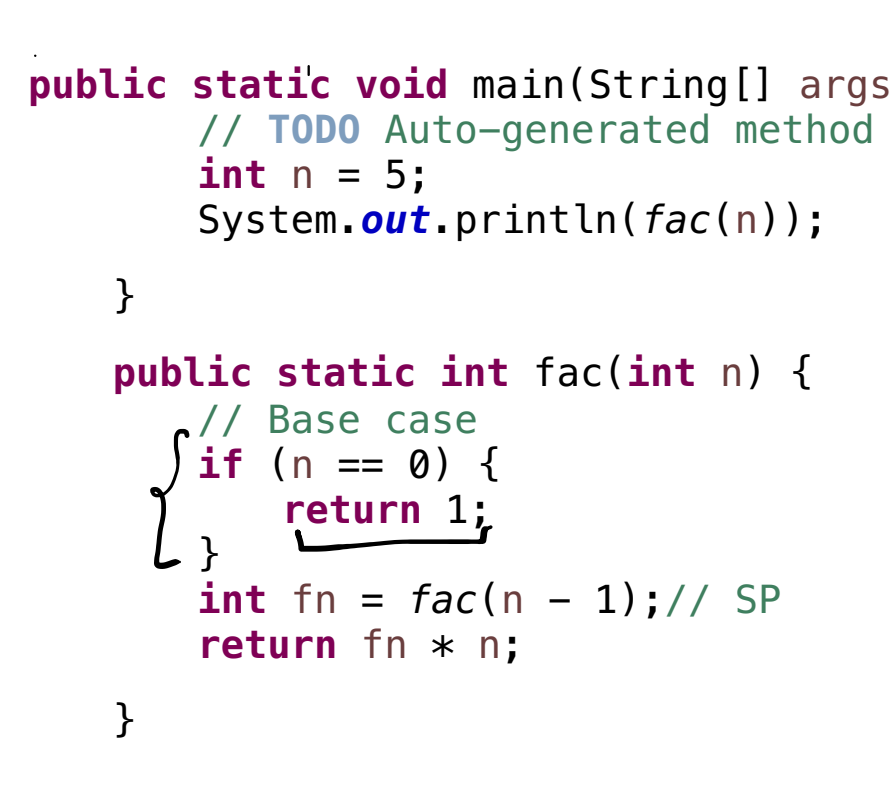
5! = 4! × 5 (24)
6 ← 3! × 4
2 ← 2! × 3
1 ← 1! × 2
← 0! × 1



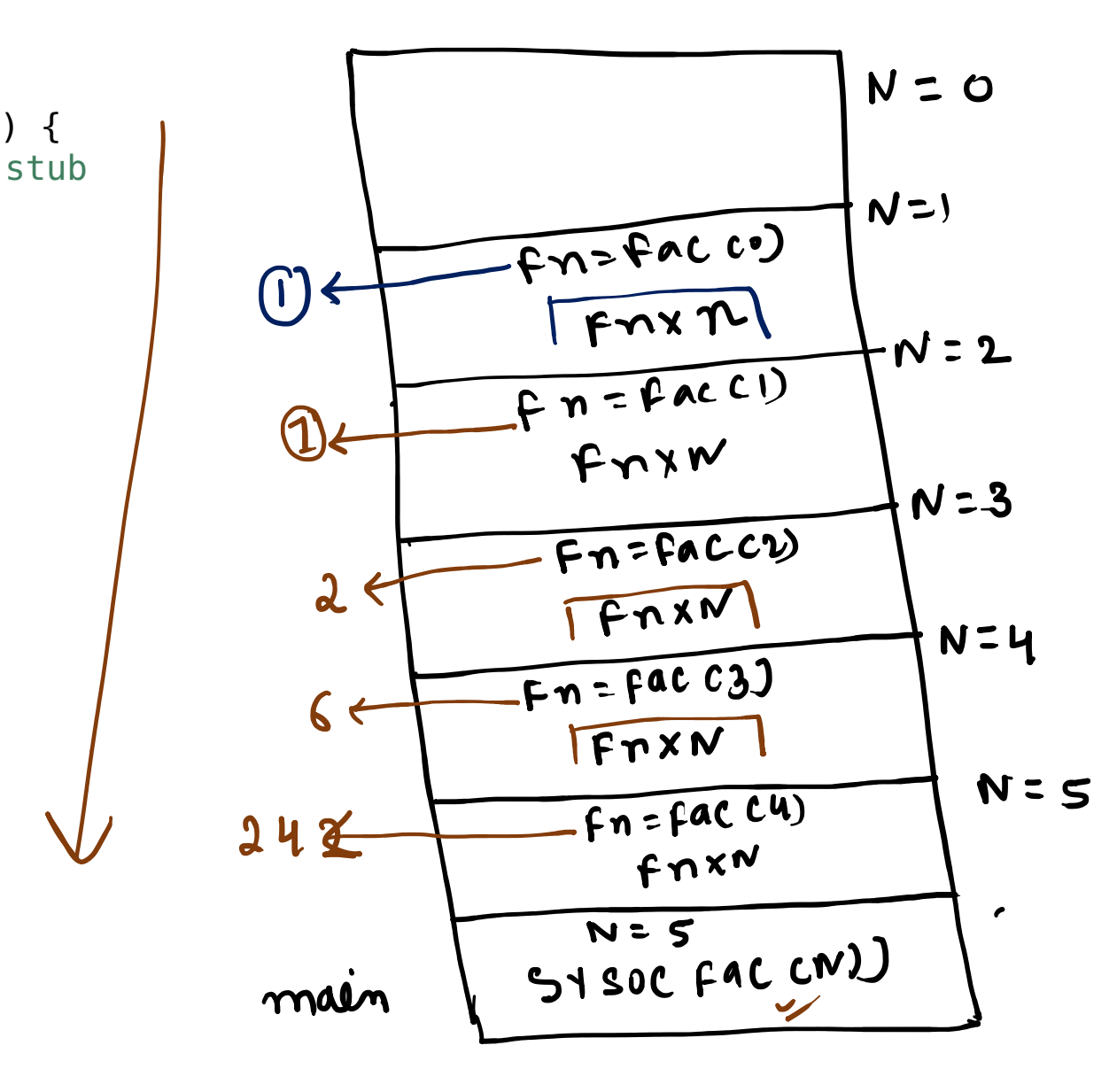
```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int n = 5;
	System.out.println(fac(n));

}
public static int fac(int n) {

	int fn = fac(n - 1);
	return fn * n;

}
```



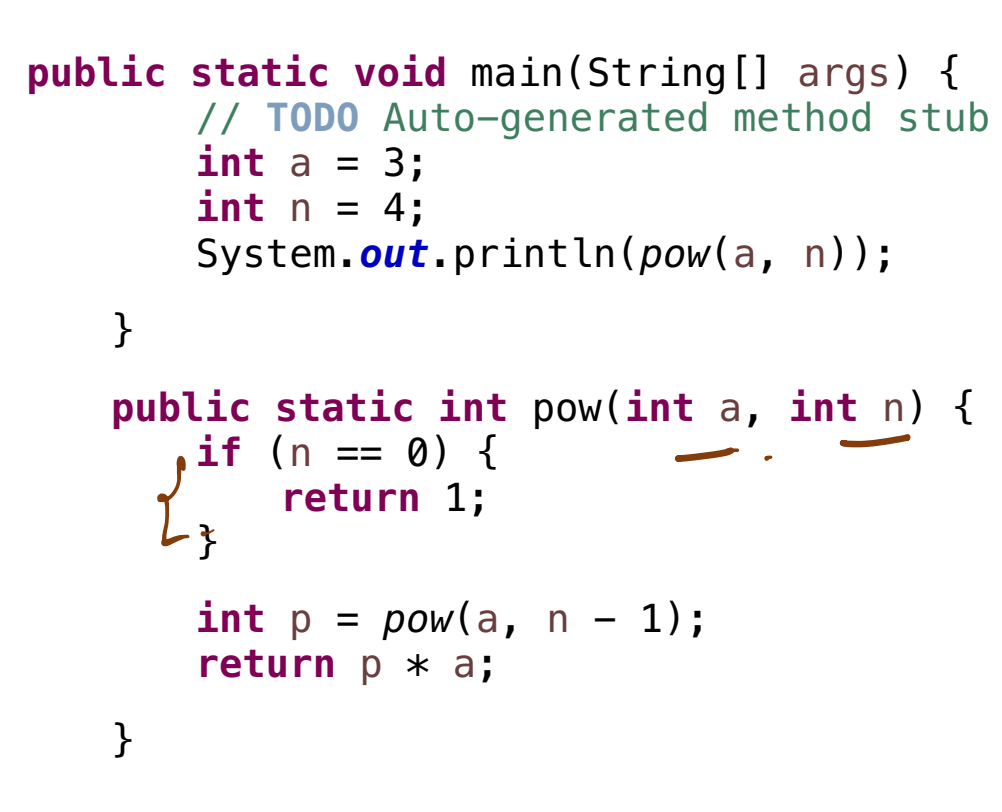
N=0 X
fn=fac(-1) fn×0
N=1
fn=fac(0) fn×1
N=2
fn=fac(1) fn×2
N=3
fn=fac(2) fn×3
N=4
fn=fac(3) fn×4
N=5
fn=fac(4) fn×N→5
main
N=5 Syso(fac(N))



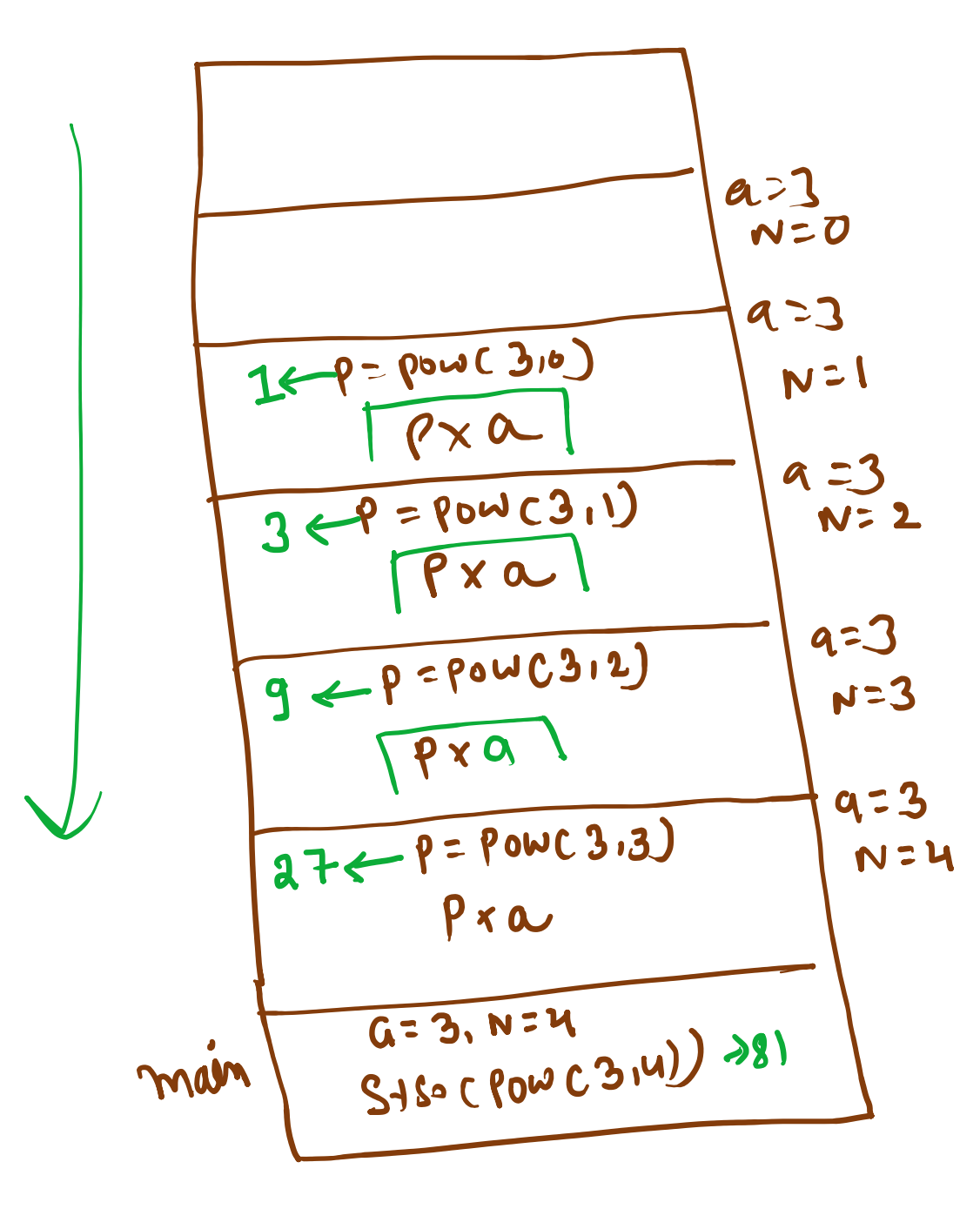
```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int n = 5;
	System.out.println(fac(n));

}

public static int fac(int n) {
	// Base case
	if (n == 0) {
		return 1;
	}
	int fn = fac(n - 1);// SP
	return fn * n;

}
```



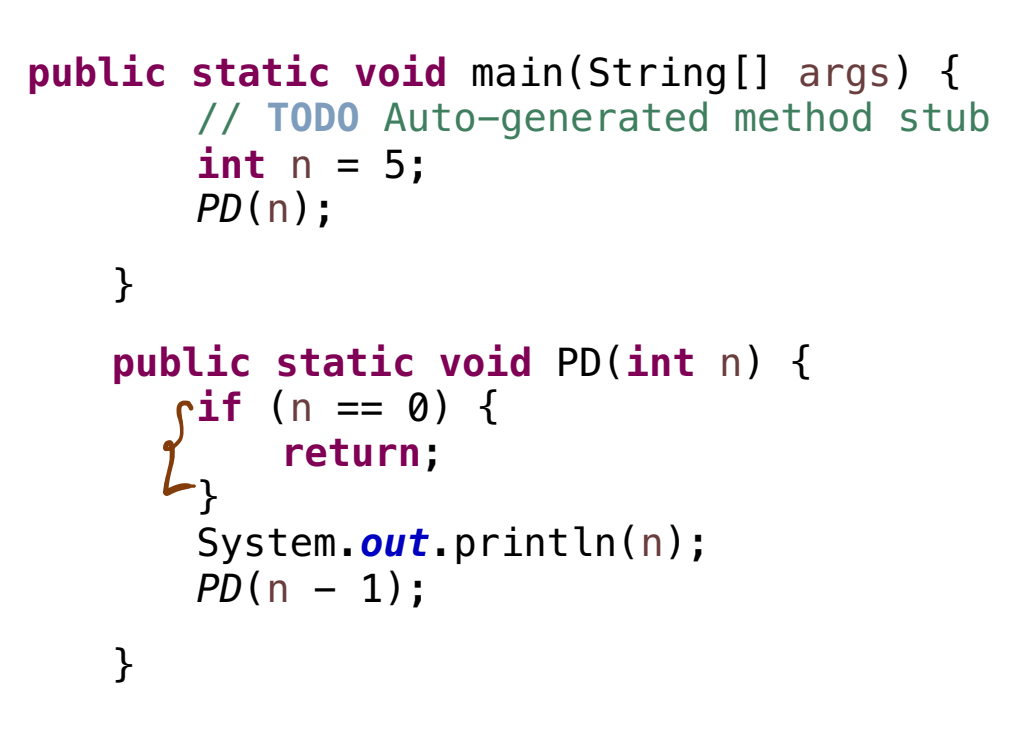
N=0
N=1
① fn=fac(0) fn×n
N=2
① fn=fac(1) fn×N
N=3
2 fn=fac(2) fn×N
N=4
6 fn=fac(3) fn×N
N=5
24 fn=fac(4) fn×N
main
N=5 Syso(fac(N))



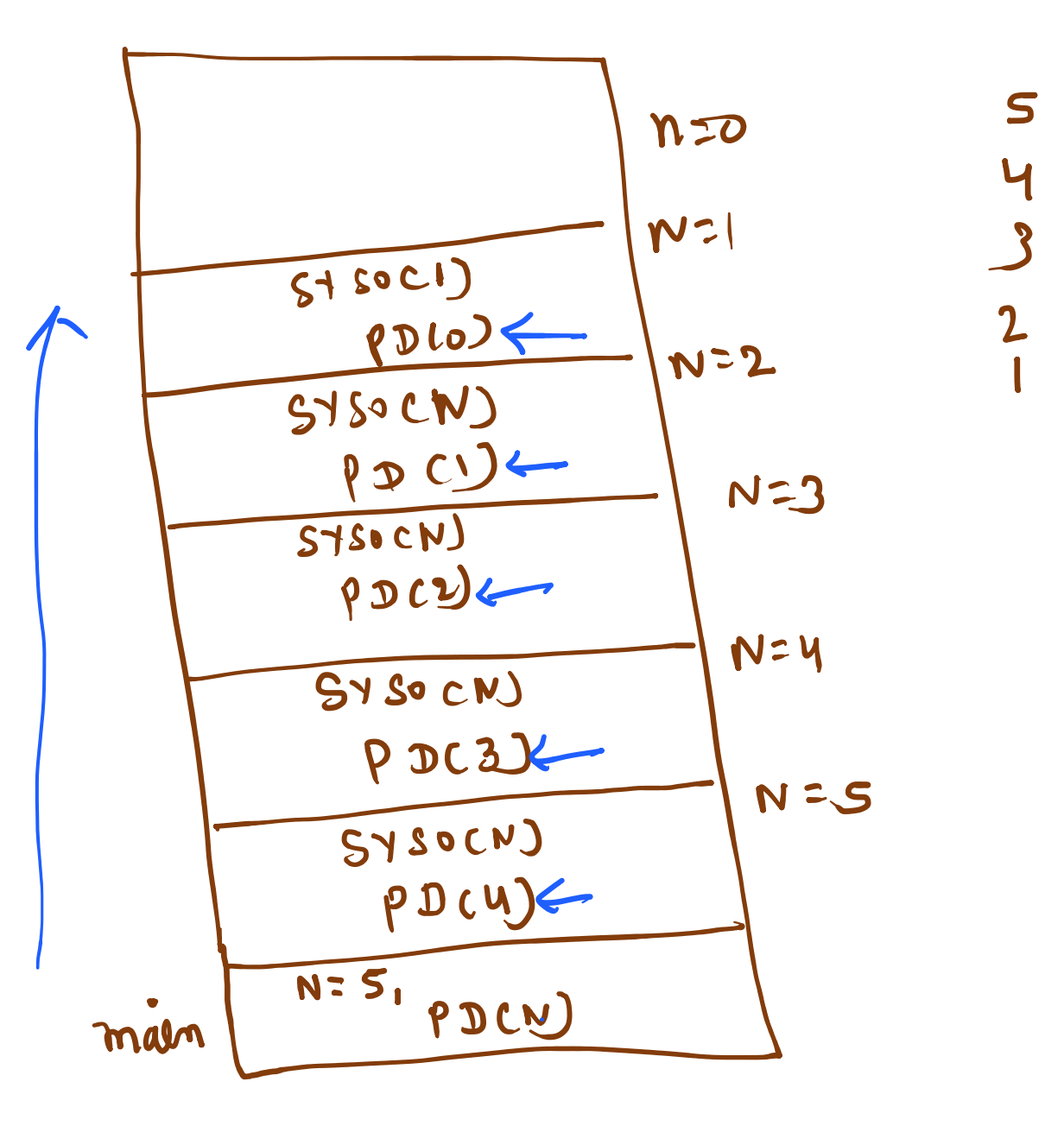
```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int a = 3;
	int n = 4;
	System.out.println(pow(a, n));

}
public static int pow(int a, int n) {
	if (n == 0) {
		return 1;
	}

	int p = pow(a, n - 1);
	return p * a;

}
```



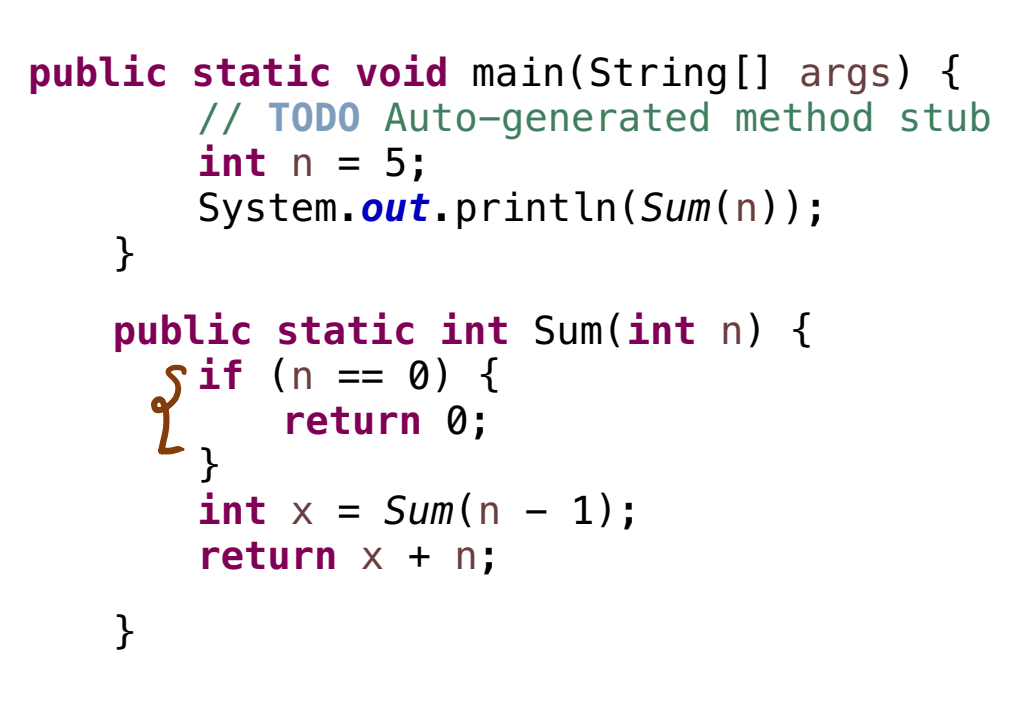
a=3 N=0
a=3 N=1
1 ← p=pow(3,0) p×a
a=3 N=2
3 ← p=pow(3,1) p×a
a=3 N=3
9 ← p=pow(3,2) p×a
a=3 N=4
27 ← p=pow(3,3) p×a
main
a=3, N=4 Syso(pow(3,4)) →81



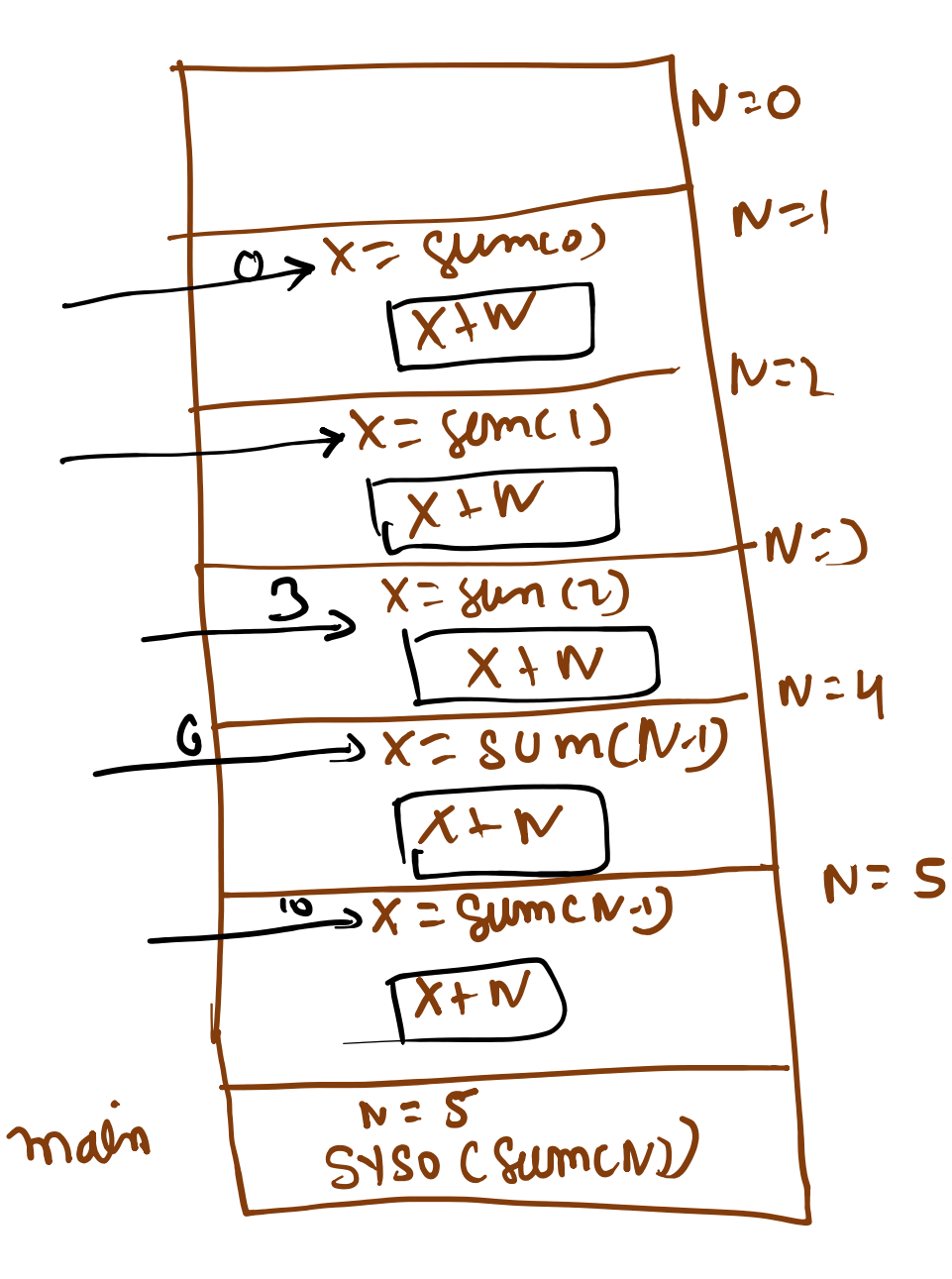
```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int n = 5;
	PD(n);

}

public static void PD(int n) {
	if (n == 0) {
		return;
	}
	System.out.println(n);
	PD(n - 1);

}
```



n=0
N=1
Syso(1) PD(0)
N=2
Syso(N) PD(1)
N=3
Syso(N) PD(2)
N=4
Syso(N) PD(3)
N=5
Syso(N) PD(4)
main
N=5, PD(N)
5
4
3
2
1



```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int n = 5;
	System.out.println(Sum(n));
}

public static int Sum(int n) {
	if (n == 0) {
		return 0;
	}
	int x = Sum(n - 1);
	return x + n;

}
```

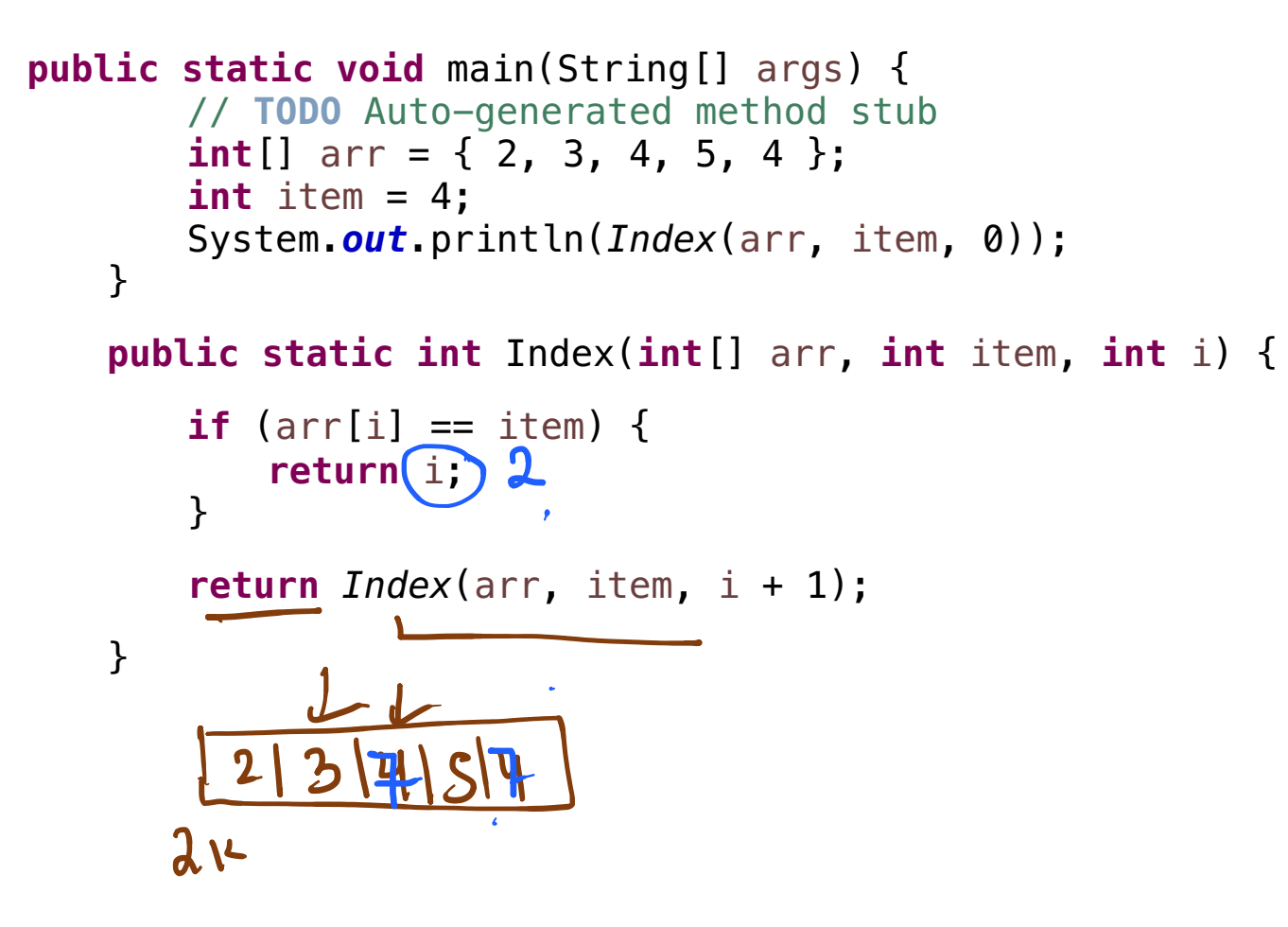


N=0
N=1
0 → X= Sum(0) X+N
N=2
X= Sum(1) X+N
N=3
3 X= Sum(2) X+N
N=4
6 X= Sum(N-1) X+N
N=5
10 X= Sum(N-1) X+N
main
N=5 Syso(Sum(N))

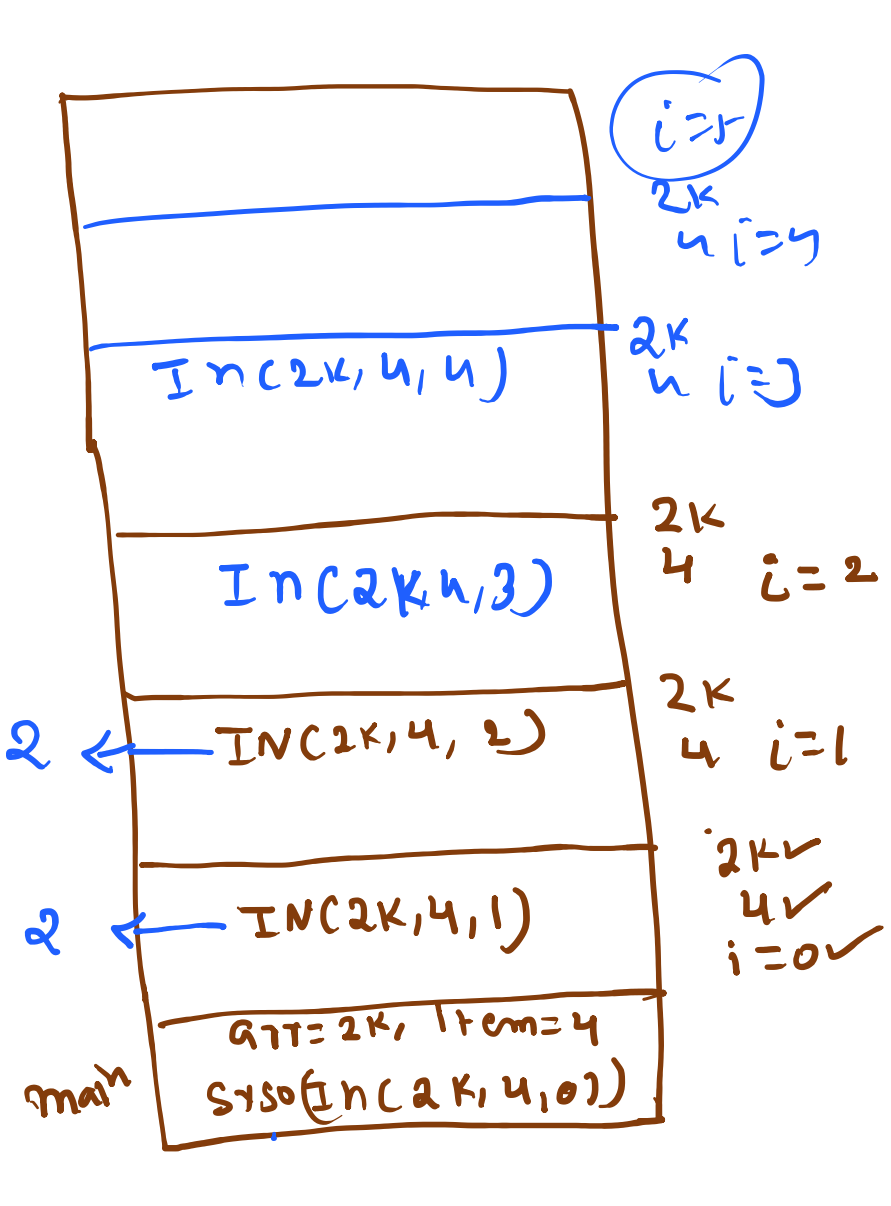
i

{ 2, 3, 4, 5, 4 };
0 1 2 3 4

if (arr(i) == item) {
ret i
}
R(arr, item, i+1)



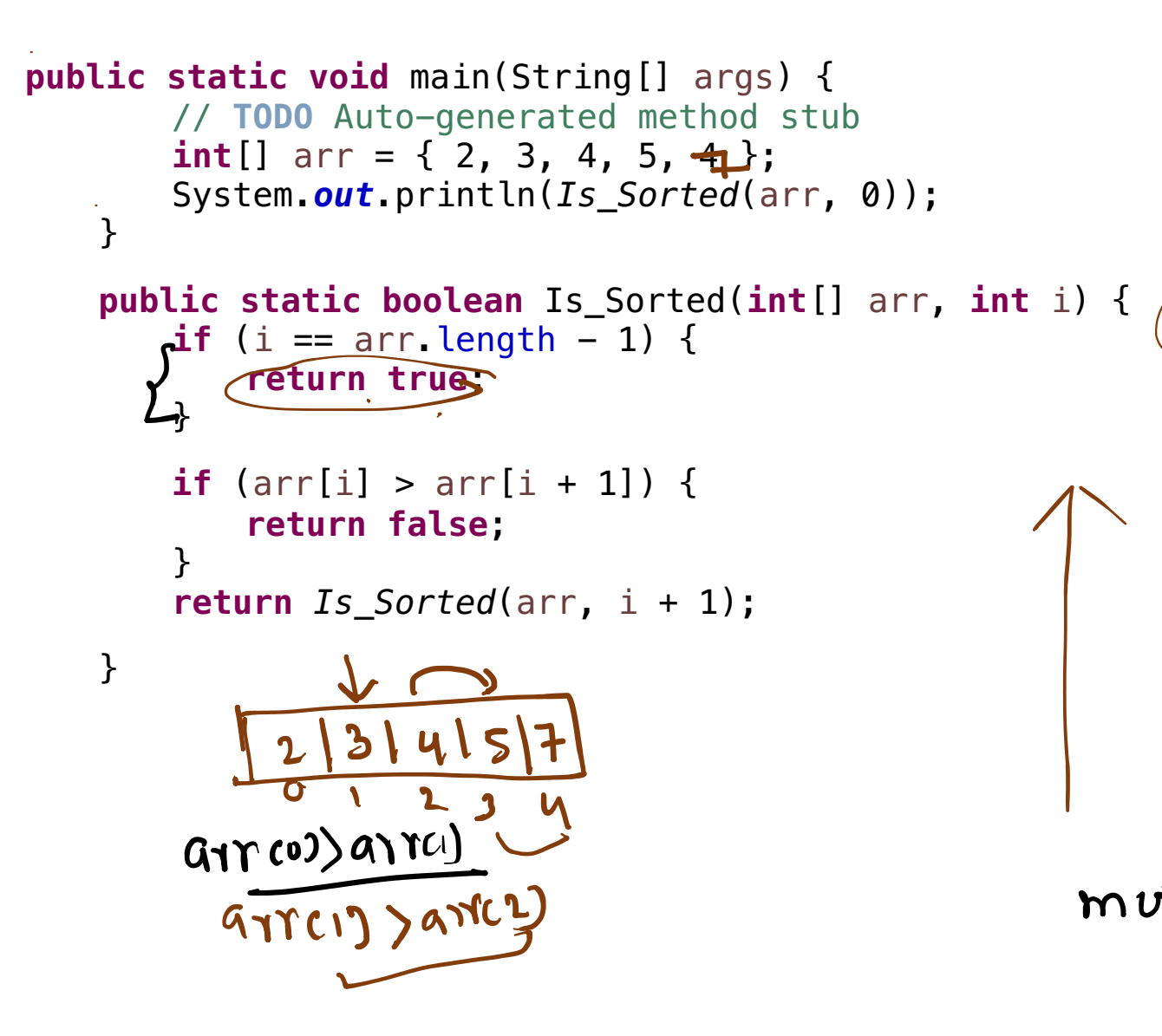
```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int[] arr = { 2, 3, 4, 5, 4 };
	int item = 4;
	System.out.println(Index(arr, item, 0));
}

public static int Index(int[] arr, int item, int i) {

	if (arr[i] == item) {
		return i;
	}

	return Index(arr, item, i + 1);

}
```



i=5
2k 4 i=4
2k 4 i=3
In(2k,4,4)
2k 4 i=2
In(2k,4,3)
2k 4 i=1
2 ← In(2k,4,2)
2k ✓ 4 ✓ i=0 ✓
2 ← In(2k,4,1)
main
arr=2k, item=4
Syso(In(2k,4,0))

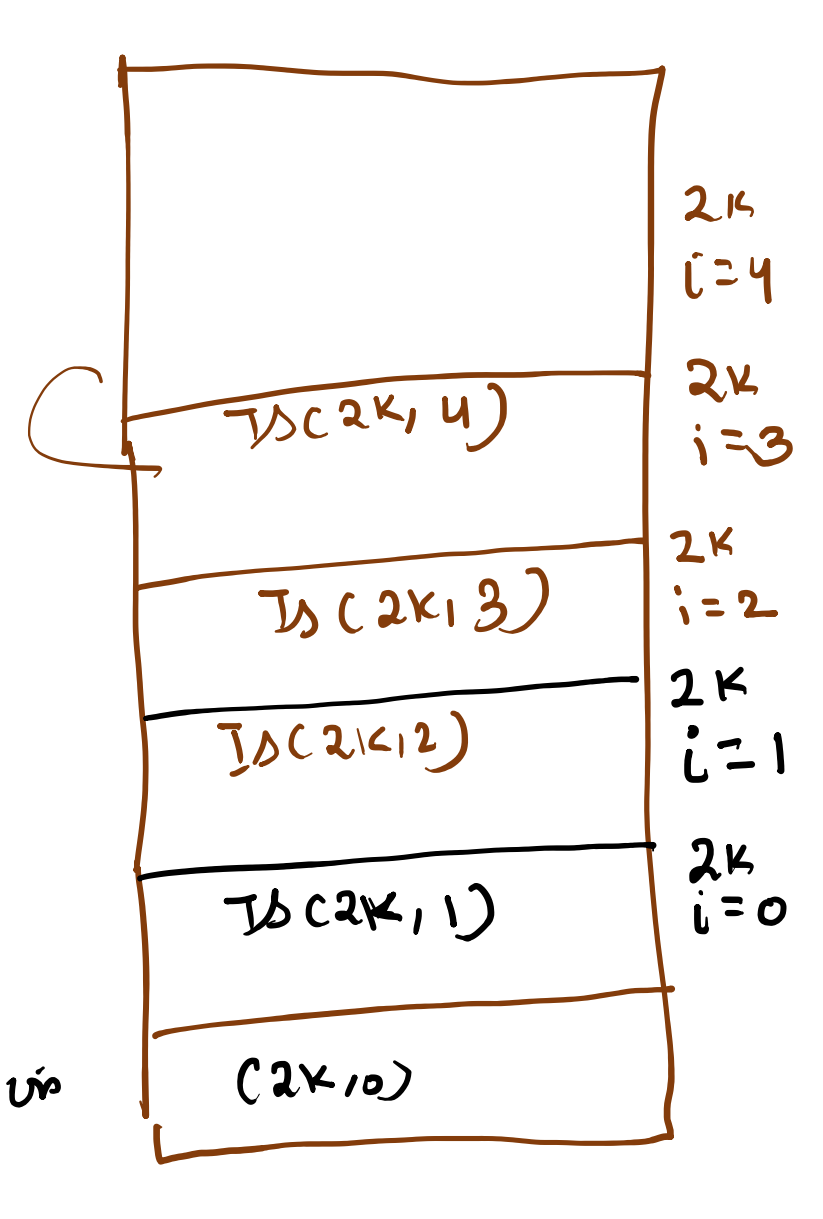
int[] arr = { 2, 3, 4, 5, 4 };
0 1 2 3 4

i=0 arr(0) > arr(1)
i=1 arr(1) > arr(2)
i=2 arr(2) > arr(3)
i=3 arr(3) > arr(4)
i=4 arr(4)

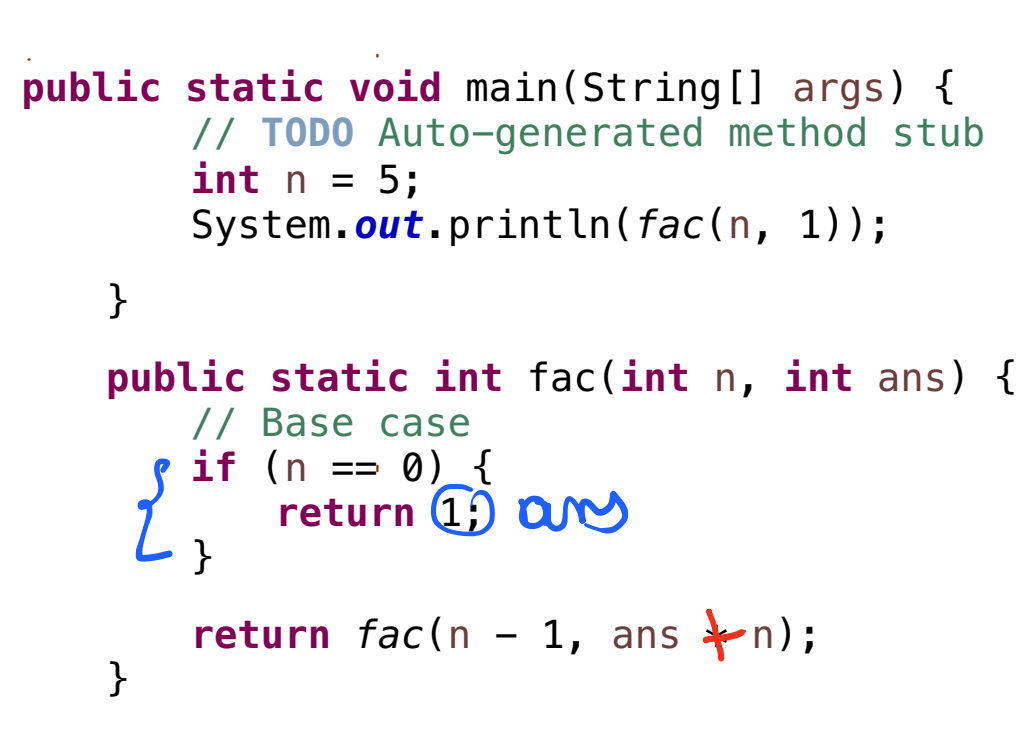


```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int[] arr = { 2, 3, 4, 5, 4 };
	System.out.println(Is_Sorted(arr, 0));
}

public static boolean Is_Sorted(int[] arr, int i) {
	if (i == arr.length - 1) {
		return true;
	}

	if (arr[i] > arr[i + 1]) {
		return false;
	}
	return Is_Sorted(arr, i + 1);

}
```

2|3|4|5|7
0 1 2 3 4
arr(0) > arr(1)
arr(1) > arr(2)
main

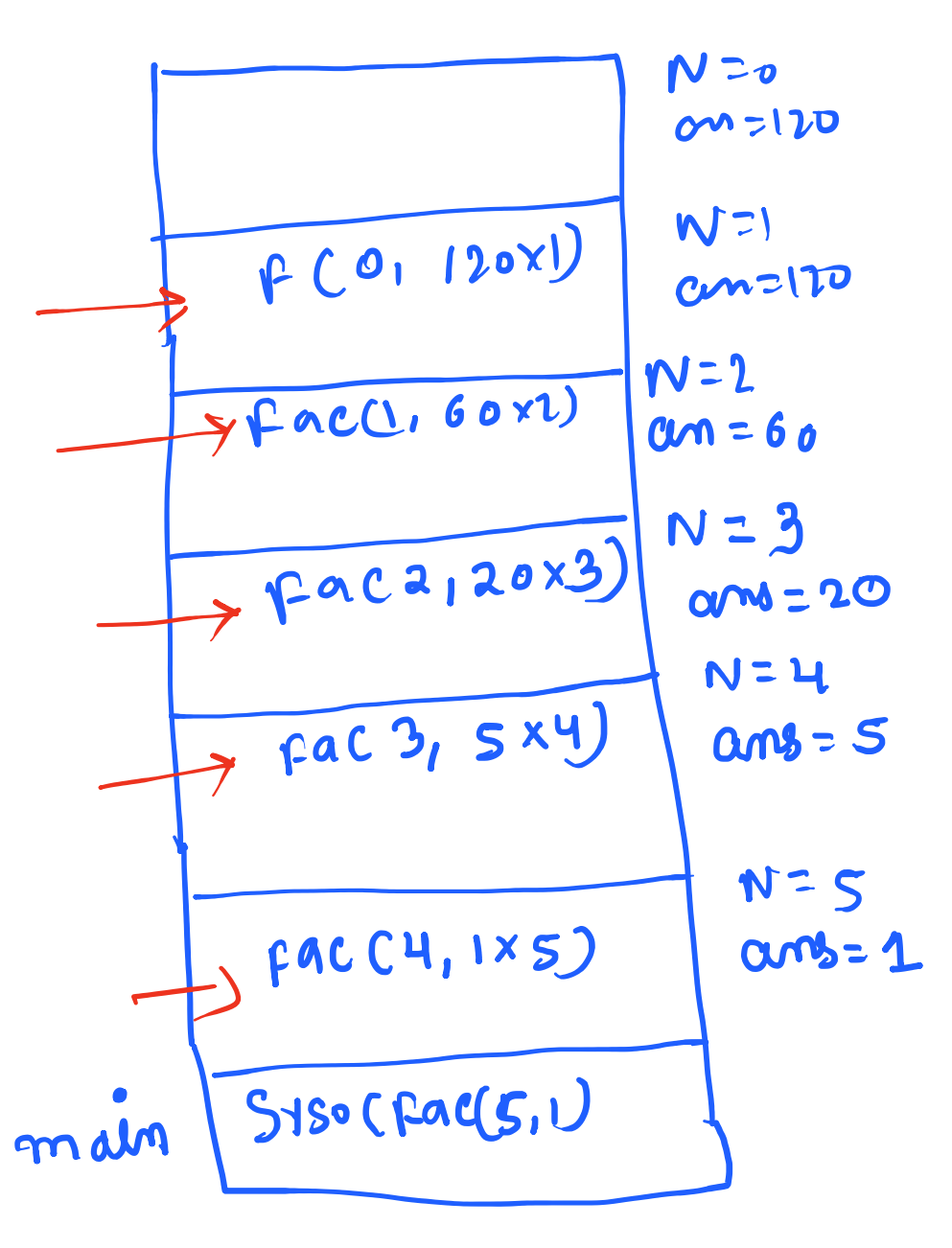


2k i=4
2k i=3
Is(2k,4)
2k i=2
Is(2k,3)
2k i=1
Is(2k,2)
2k i=0
Is(2k,1)
(2k,0)



```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int n = 5;
	System.out.println(fac(n, 1));

}

public static int fac(int n, int ans) {
	// Base case
	if (n == 0) {
		return 1;
	}

	return fac(n - 1, ans + n);
}
```

pow(a, n-1, ans×a)

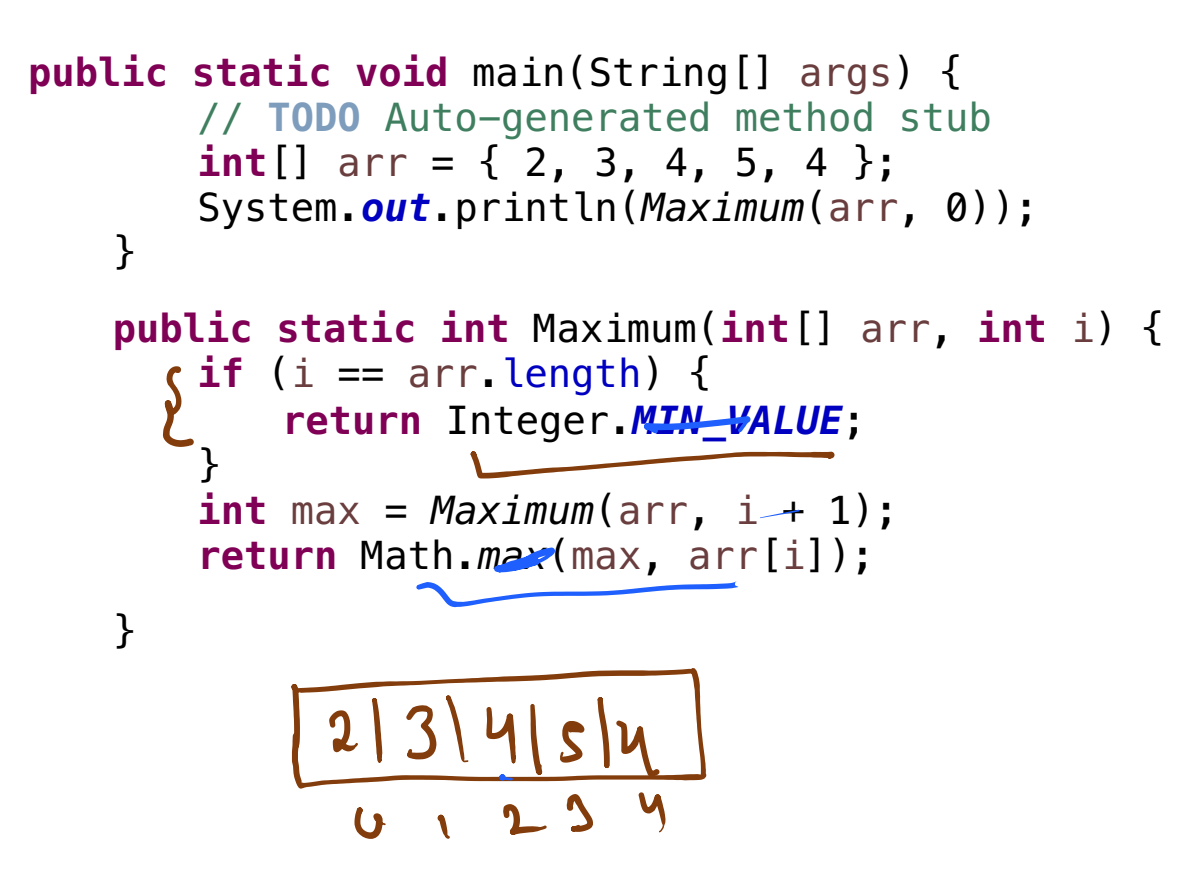


N=0 ans=120
N=1 ans=120
f(0, 120×1)
N=2 ans=60
fac(1, 60×2)
N=3 ans=20
fac(2, 20×3)
N=4 ans=5
fac(3, 5×4)
N=5 ans=1
fac(4, 1×5)
main
Syso(fac(5,1))

i= = arr.len-

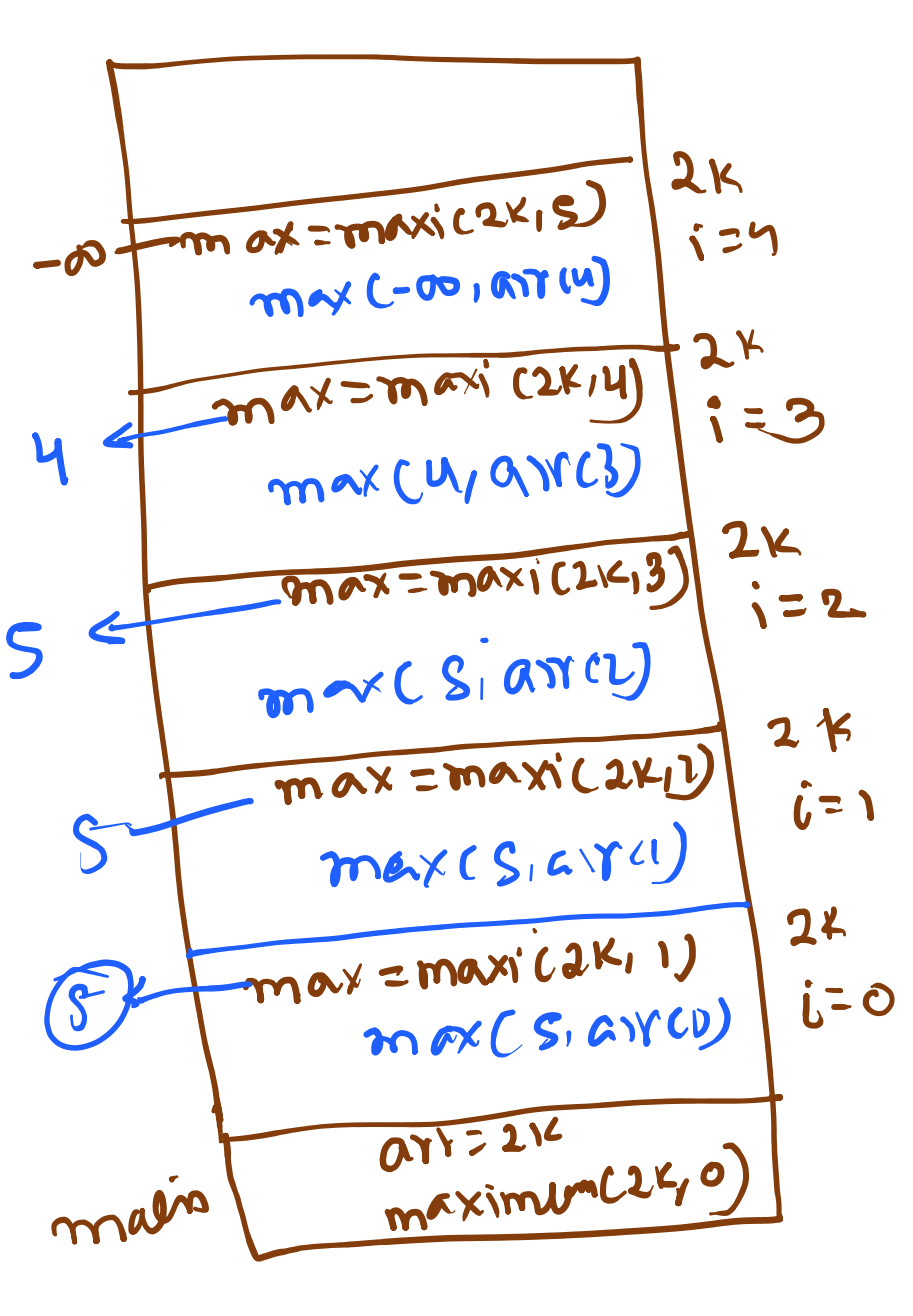
2, 3, 4, 5, 4
0 1 2 3 4
i

int max= R(arr, i+1)
Math.max(arr(i), max)

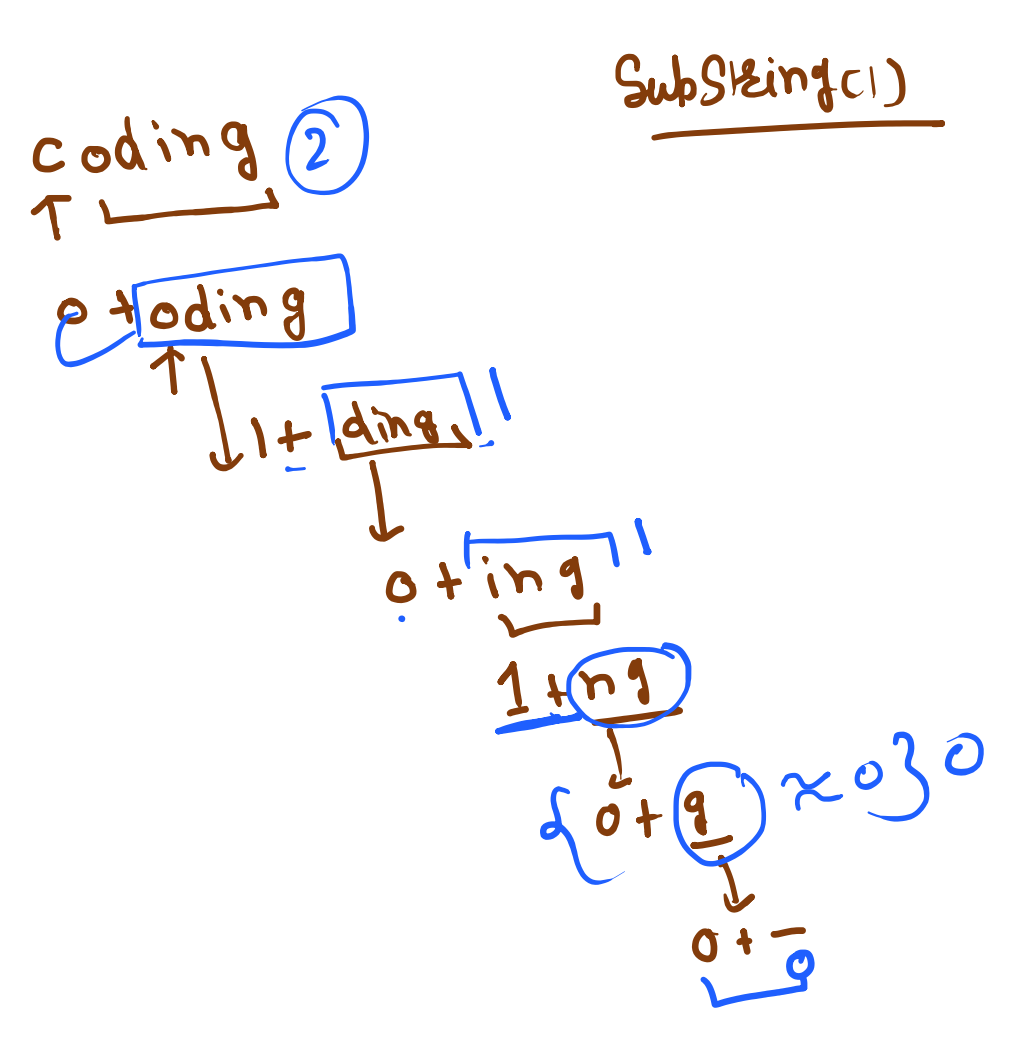


```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int[] arr = { 2, 3, 4, 5, 4 };
	System.out.println(Maximum(arr, 0));
}

public static int Maximum(int[] arr, int i) {
	if (i == arr.length) {
		return Integer.MIN_VALUE;
	}
	int max = Maximum(arr, i + 1);
	return Math.max(max, arr[i]);

}
```

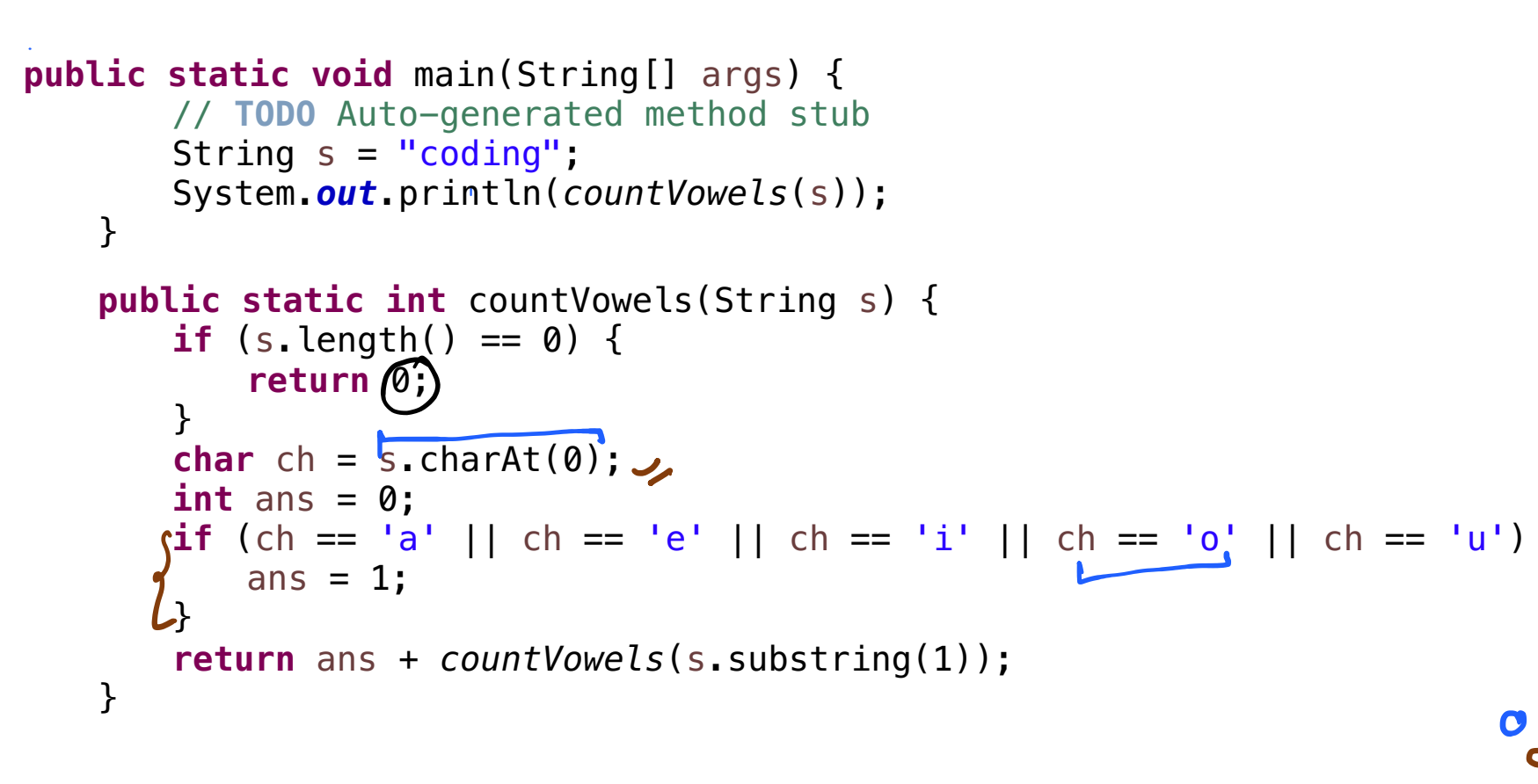
2|3|4|5|4
0 1 2 3 4



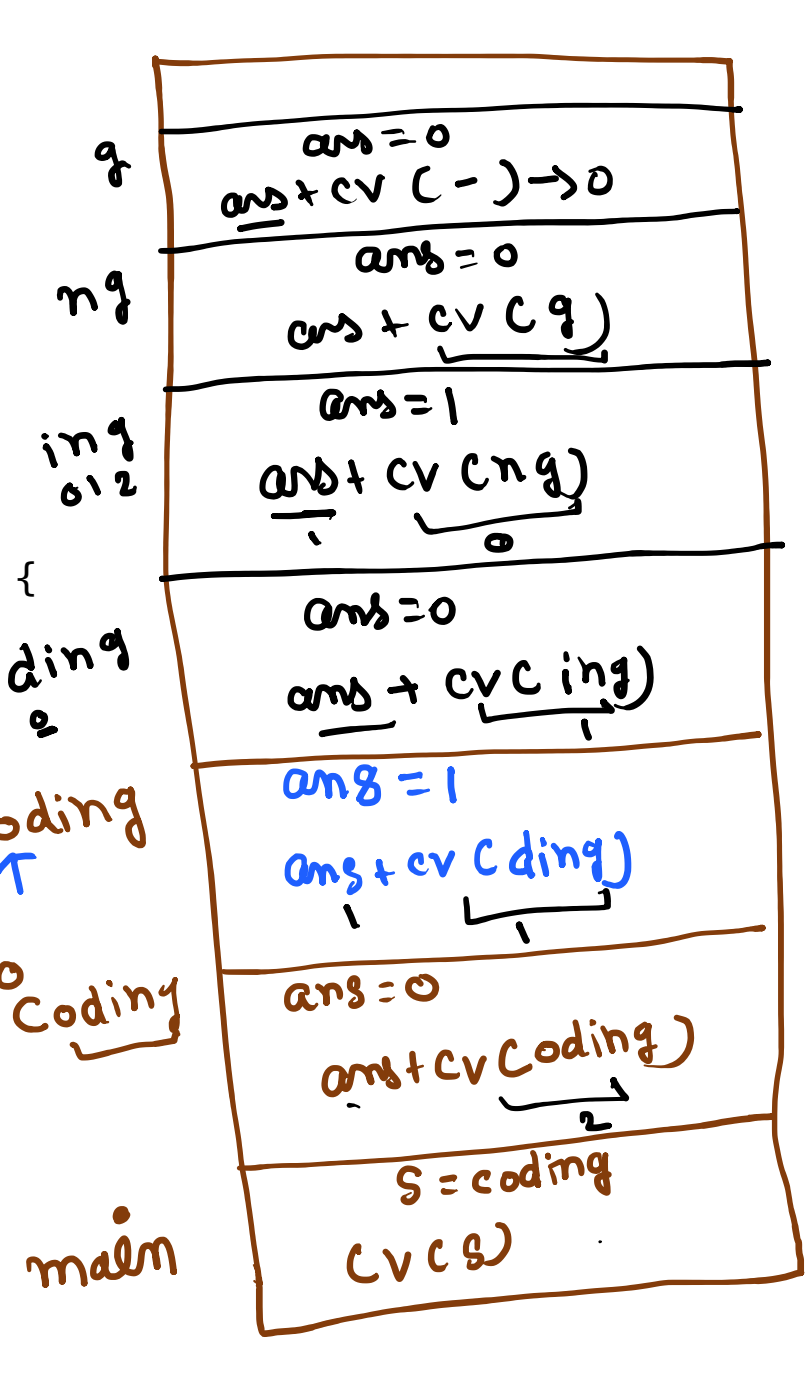
2k i=4
-∞ max=max(2k,5) max(-∞, arr(4))
2k i=3
4 max=max(2k,4) max(4, arr(3))
2k i=2
5 max=max(2k,3) max(5, arr(2))
2k i=1
5 max=max(2k,2) max(5, arr(1))
2k i=0
5 max=max(2k,1) max(5, arr(0))
main
arr=2k maximum(2k,0)



Substring(1)
coding 2
0 + oding
1 + ding 1
0 + ing 1
1 + ng
0 + g ×0}0
0 + –
0



```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	String s = "coding";
	System.out.println(countVowels(s));
}

public static int countVowels(String s) {
	if (s.length() == 0) {
		return 0;
	}
	char ch = s.charAt(0);
	int ans = 0;
	if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
		ans = 1;
	}
	return ans + countVowels(s.substring(1));
}
```



g
ans=0
ans+cv(–)→0
ng
ans=0
ans+cv(g)
ing 012
ans=1
ans+cv(ng)
ding 2
ans=0
ans+cv(ing)
oding
ans=1
ans+cv(ding)
coding
ans=0
ans+cv(coding)
main
S=coding
cv(S)

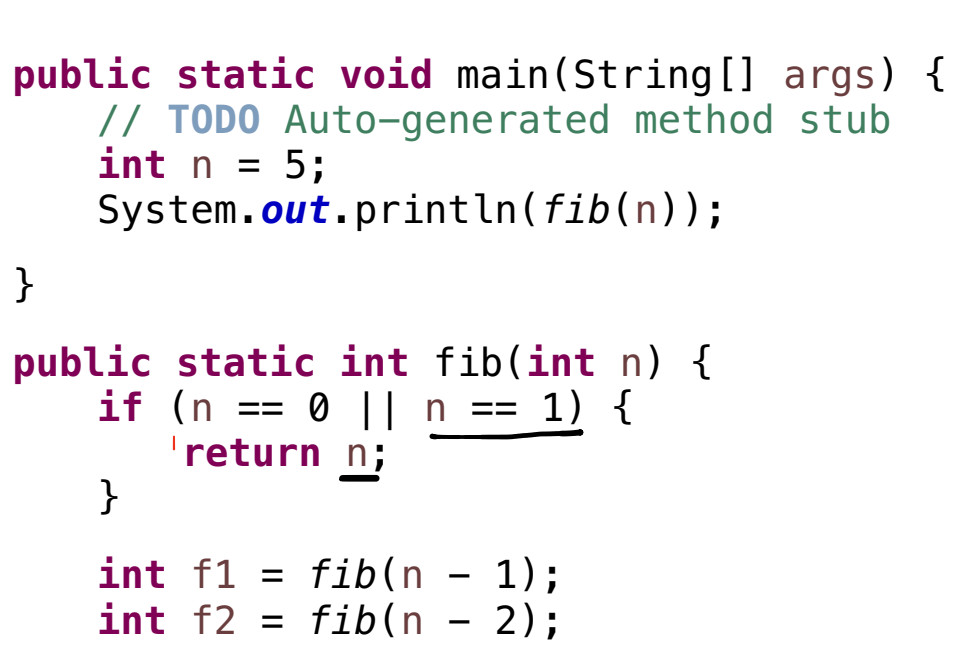
Fib(n)

fib(n-1) + fib(n-2)
Fn

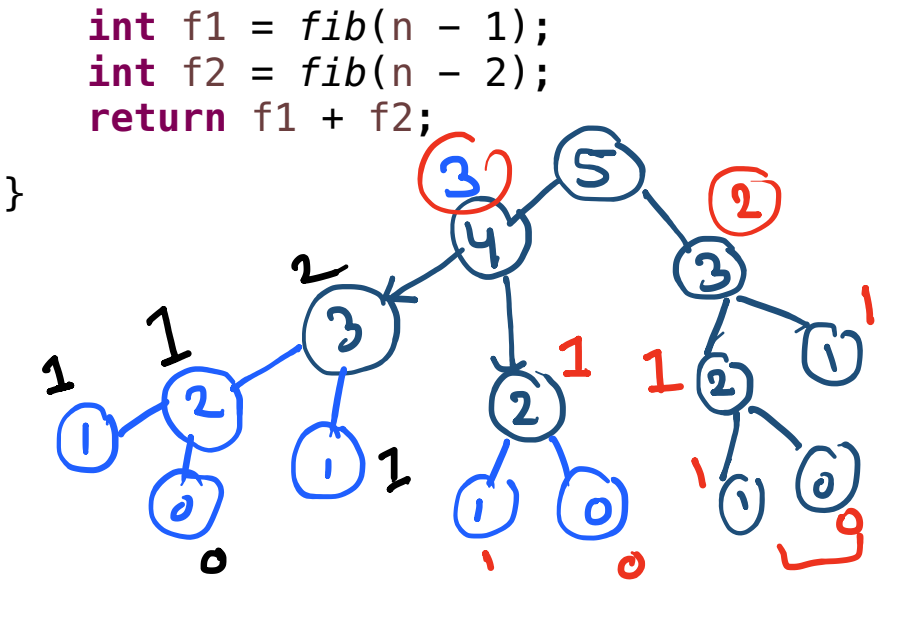
x→( )

2016 [illegible]

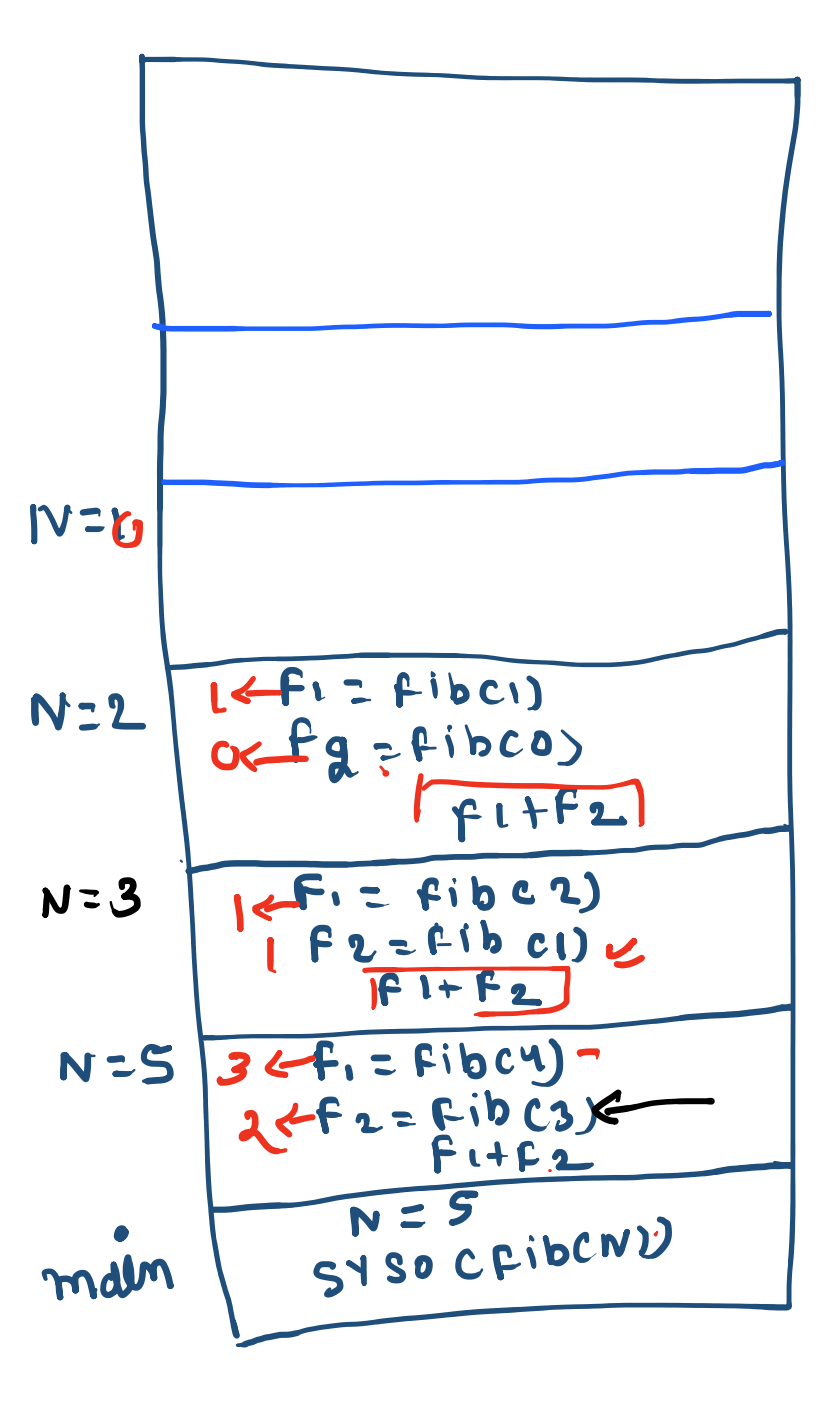
fib(int n){
n==0 || n==1
ret n
f1 = fib(n-1) ✓
f2 = fib(n-2) ✓
return f1+f2
}



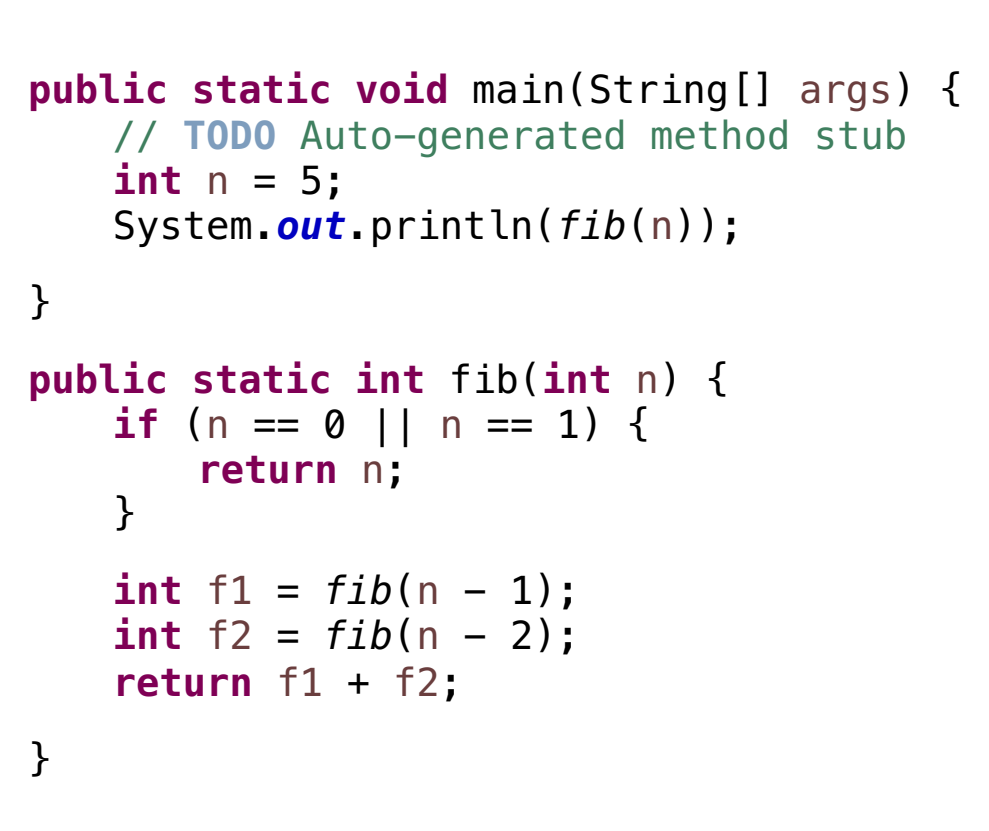
```java
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    System.out.println(fib(n));
}

public static int fib(int n) {
    if (n == 0 || n == 1) {
        return n;
    }

    int f1 = fib(n - 1);
    int f2 = fib(n - 2);
    return f1 + f2;
}
```



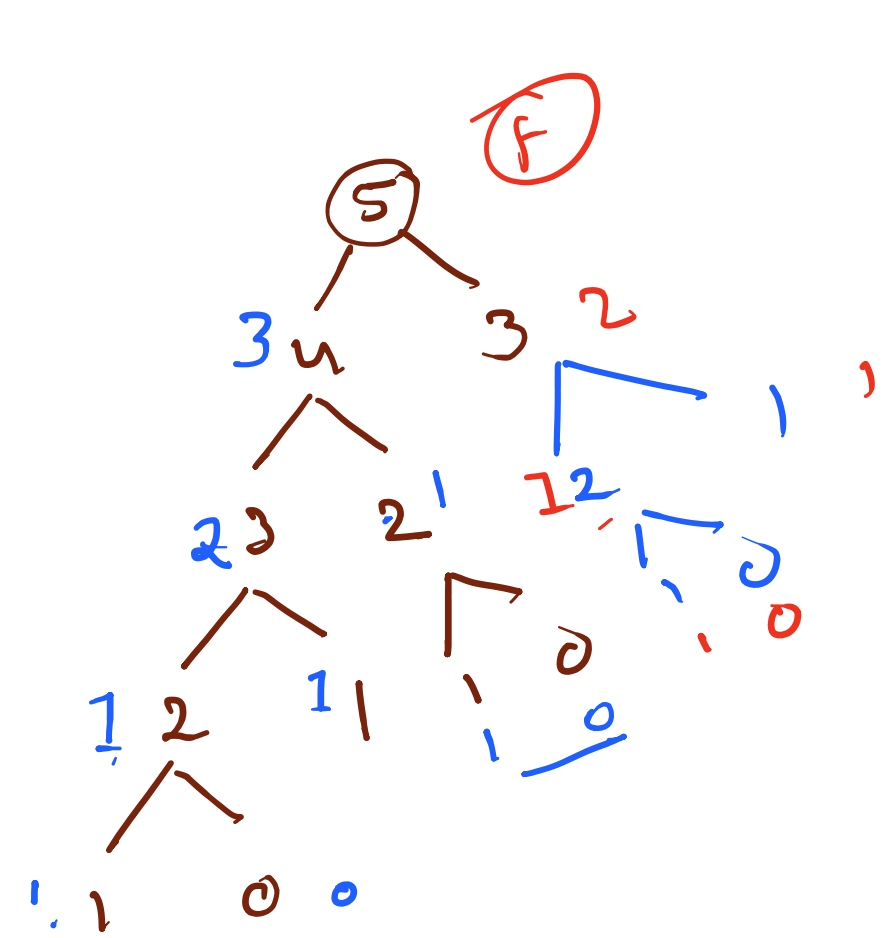
5
3
4
2
3
3
1
1
2
1
1
2
0
0
1
1
1
2
1
0
1
0
1
0



N=0
N=2
1 ← f1 = fib(1)
0 ← f2 = fib(0)
f1+f2
N=3
1 ← f1 = fib(2)
1 f2 = fib(1)
f1+f2
N=5
3 ← f1 = fib(4)
2 ← f2 = fib(3)
f1+f2
main
N=5
Syso(fib(N))



```java
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    System.out.println(fib(n));

}

public static int fib(int n) {
    if (n == 0 || n == 1) {
        return n;
    }

    int f1 = fib(n - 1);
    int f2 = fib(n - 2);
    return f1 + f2;

}
```



F
5
3
4
3
2
1
1
2
3
2
1
1
2
1
0
0
1
2
1
1
0
1
0
1
1
0
0